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LABORATORY REPORT NO. 2

Cryptanalysis of Monoalphabetic Ciphers

Cryptography: Variant 12, frequency analysis and substitution recovery

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ABSTRACT

This laboratory examines the recovery of an English message encrypted with a monoalphabetic substitution cipher for Variant 12. Letter frequencies provide initial hypotheses, while short recurring words, time expressions, common letter combinations and grammatical context constrain the remaining substitutions. The cryptogram contains 2,725 alphabetic characters and 24 distinct ciphertext symbols. All observed symbols are resolved consistently through 12 progressive deduction stages, and the two unused ciphertext letters are paired with the remaining unused plaintext letters by elimination. Intermediate extracts and frequency table captures illustrate the gradual recovery of the passage about the Viennese Black Chamber (*Geheime Kabinettskanzlei*), and the complete ciphertext and literal plaintext are preserved in the appendices.

Keywords: cryptanalysis, monoalphabetic substitution, frequency analysis, word patterns, plaintext recovery

INTRODUCTION

The objective of this laboratory work is to decrypt the English monoalphabetic cryptogram assigned for Variant 12 using frequency analysis and linguistic pattern recognition [1]. Only alphabetic characters are encrypted; spaces, digits and punctuation remain intact and provide structural clues during cryptanalysis.

1 SHORT THEORY

A fixed substitution changes the labels of letters while preserving the statistical and structural patterns that make a natural language recognizable.

1.1 Monoalphabetic substitution

Let A be the 26-letter English alphabet and let π be a permutation of A. For plaintext letter p_i and ciphertext letter c_i, encryption and decryption are:

$$c_i = \pi(p_i), \qquad p_i = \pi^{-1}(c_i). \tag{1.1}$$

The same plaintext letter always produces the same ciphertext symbol. Consequently, letter counts, repeated-letter patterns and word lengths survive encryption. A general substitution has $26! \approx 4.03 \times 10^{26}$ possible keys, making exhaustive key search infeasible, but frequency analysis recovers the permutation directly from statistical and lexical properties of the ciphertext.

1.2 Letter frequencies

For a ciphertext of N alphabetic characters, the relative frequency of symbol c is:

$$f_C(c) = \frac{n_C(c)}{N} \times 100\%, \tag{1.2}$$

where $n_C(c)$ denotes its count. Uppercase and lowercase letters are counted together; digits, punctuation, spaces and line breaks are excluded. Table 1.1 lists the standard reference frequencies of English letters, alongside the observed counts and percentages in Variant 12 ($N = 2{,}725$).

Table 1.1 – Standard English letter frequencies and observed Variant 12 ciphertext distribution ($N = 2{,}725$)

English	Ref. (%)	Ciphertext (ranked)	Count	Observed (%)	Alphabetical (A–Z)
E	12.70	V	380	13.94	A: 45 (1.65%)
T	9.06	W	274	10.06	B: 0 (0.00%)
A	8.17	T	227	8.33	C: 48 (1.76%)
O	7.51	N	185	6.79	D: 64 (2.35%)
I	6.97	P	185	6.79	E: 3 (0.11%)
N	6.75	G	176	6.46	F: 42 (1.54%)
S	6.33	X	174	6.39	G: 176 (6.46%)
H	6.09	I	170	6.24	H: 83 (3.05%)
R	5.99	Q	142	5.21	I: 170 (6.24%)
D	4.25	S	120	4.40	J: 51 (1.87%)
L	4.03	O	111	4.07	K: 19 (0.70%)
C	2.78	H	83	3.05	L: 21 (0.77%)
U	2.76	U	68	2.50	M: 0 (0.00%)
M	2.41	Z	68	2.50	N: 185 (6.79%)
W	2.36	R	65	2.39	O: 111 (4.07%)
F	2.23	D	64	2.35	P: 185 (6.79%)
G	2.01	J	51	1.87	Q: 142 (5.21%)
Y	1.97	C	48	1.76	R: 65 (2.39%)
P	1.93	A	45	1.65	S: 120 (4.40%)
B	1.49	F	42	1.54	T: 227 (8.33%)
V	0.98	L	21	0.77	U: 68 (2.50%)
K	0.77	K	19	0.70	V: 380 (13.94%)
J	0.15	Y	4	0.15	W: 274 (10.06%)
X	0.15	E	3	0.11	X: 174 (6.39%)
Q	0.09	B	0	0.00	Y: 4 (0.15%)
Z	0.07	M	0	0.00	Z: 68 (2.50%)

1.3 Word patterns and consistency

Common English digraphs include `TH`, `HE`, `AN`, `IN`, `ER` and `ON`, while common trigraphs include `THE`, `AND`, `THA`, `ENT` and `ION`. One-letter words (*a* and *I*), doubled letters, time notations such as `a.m.` and `p.m.`, and repeated short words constrain the substitution key. Because π is a bijection, each ciphertext symbol maps to a unique plaintext letter, and every tentative replacement must remain consistent across the entire message.

2 PRACTICAL ANALYSIS

All mappings below run from ciphertext to plaintext: $V \rightarrow e$ means that ciphertext V represents plaintext e. In intermediate decryptions, unresolved ciphertext symbols remain uppercase and resolved plaintext letters are written in lowercase.

2.1 Initial frequency distribution

The Variant 12 cryptogram contains $2{,}725$ letters and 24 distinct ciphertext symbols; B and M do not appear. Figure 2.1 shows the initial ranked frequency distribution of the intercept compared with standard English letter frequencies.

E	T	A	O	I	N	S	H	R	D	L	C	U	M	W	F	G	Y	P	B	V	K	J	X	Q	Z
12.7	9.1	8.2	7.5	7.0	6.7	6.3	6.1	6.0	4.3	4.0	2.8	2.8	2.4	2.4	2.2	2.0	2.0	1.9	1.5	1.0	0.8	0.15	0.15	0.10	0.07

The frequencies of the intercept are:

V	W	T	N	P	G	X	I	Q	S	O	H	U	Z	R	D	J	C	A	F	L	K	Y	E	B	M
380	274	227	185	185	176	174	170	142	120	111	83	68	68	65	64	51	48	45	42	21	19	4	3	0	0
13.9	10.1	8.3	6.8	6.8	6.5	6.4	6.2	5.2	4.4	4.1	3.0	2.5	2.5	2.4	2.3	1.9	1.8	1.7	1.5	0.8	0.7	0.1	0.1	0.0	0.0

Figure 2.1 – Initial frequency distribution of Variant 12 compared with standard English frequencies

Ciphertext V occurs 380 times (13.94%, displayed as 13.9%) and W occurs 274 times (10.06%, displayed as 10.1%), closely matching the leading English letters e (12.7%) and t (9.1%). Ciphertext T is third with 227 occurrences (8.33%), while N and P are tied at 185 occurrences (6.79%).

Figures 2.2 and 2.3 compare the alphabetical frequency profile of standard English with the alphabetical frequency profile of the Variant 12 ciphertext. The prominent peaks at E and T in English correspond to the peaks at V and W in the encrypted text.

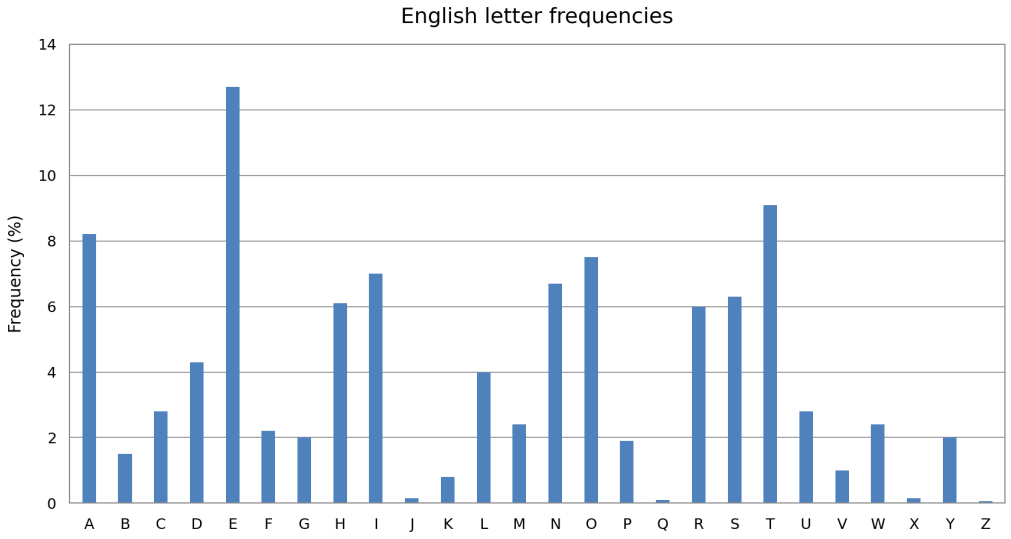


Figure 2.2 – Alphabetical frequency profile of standard English

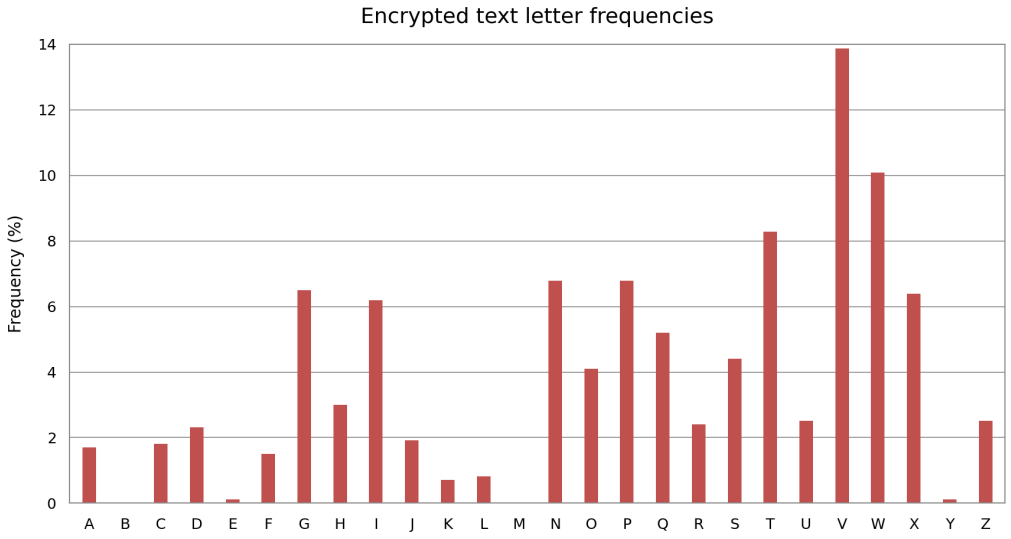


Figure 2.3 – Alphabetical frequency profile of the Variant 12 ciphertext

2.2 Progressive substitution steps

2.2.1 Step 1: Substituting $V \to e$ and $W \to t$

Based on the two highest frequency peaks (`V` at 13.9% and `W` at 10.1%), `V` is replaced with `e` and `W` with `t` (Figure 2.4).

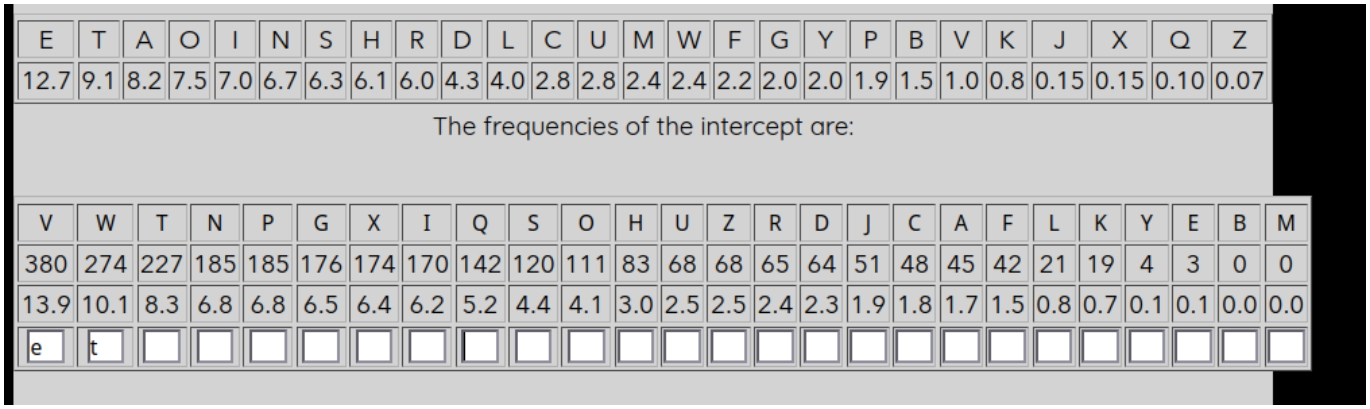


Figure 2.4 – Substitution of $V \to e$ and $W \to t$

```
Xt ITG RXtQ TSZNPt DGAeSXeKTASe eCCXHXeGHF. tQe ATJP NC ZTXS CNI OeSXKeIFtQTt ZNIGXGJ
    tN tQe
eZATPPXeP XG KXeGGT ReIe AINDJQt tN tQe ASTHLHQTZAeI eTHQ OTF Tt 7 T.Z. tQeIe tQe
   SetteIP ReIe
NUeGeO AF ZeStXGJtQeXI PeTSP RXtQ T HTGOSe. tQe NIOeI NC tQe SetteIP XG TG eGKeSNUe
   RTPGNteO
TGO tQe SetteIP JXKeG tN T PDAOXIeHtNI. Qe IeTO tQeZ TGO NIOeIeOtQe XZUNItTGt UTItP
   HNUXeO.
```

2.2.2 Step 2: Substituting $Q \to h$

The partial decryption exposes repeated occurrences of the three-letter word `tQe` and the pattern `tQTt`. Setting $Q \to h$ (142 occurrences, 5.2%) turns `tQe` into `the` and `tQTt` into `thTt` (Figure 2.5).

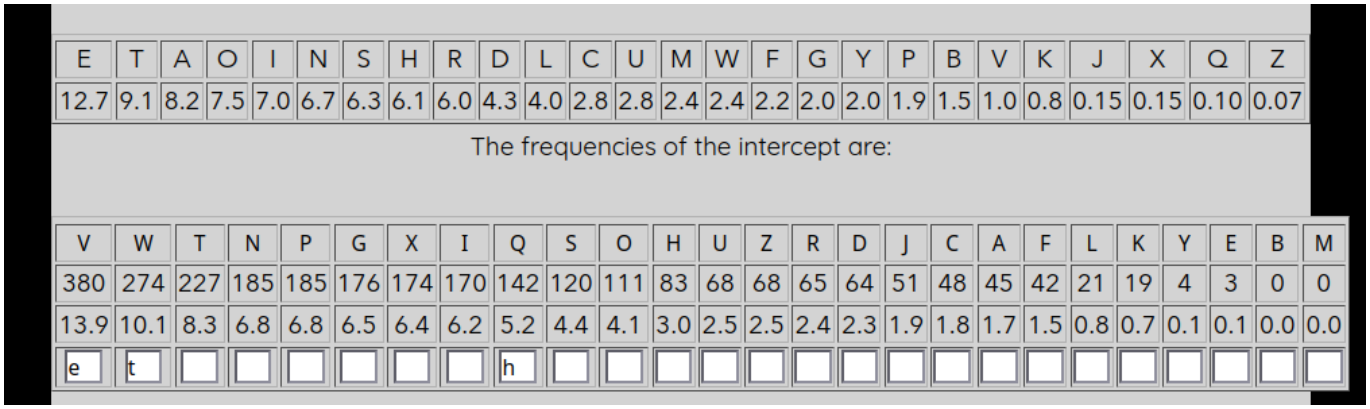


Figure 2.5 – Substitution of $Q \to h$ to form `the`

```
Xt ITG RXth TSZNPt DGAeSXeKTASe eCCXHXeGHF. the ATJP NC ZTXS CNI OeSXKeIFthTt ZNIGXGJ
    tN the
eZATPPXeP XG KXeGGT ReIe AINDJht tN the ASTHLHhTZAeI eTHh OTF Tt 7 T.Z. theIe the
   SetteIP ReIe
```

NUeGeO AF ZeStXGJtheXI PeTSP RXth T HTGOSe. the NIOeI NC the SetteIP XG TG eGKeSNUe RTPGNteO
TGO the SetteIP JXKeG tN T PDAOXIeHtNI. he IeTO theZ TGO NIOeIeOthe XZUNItTGt UTItP HNUXeO.

2.2.3 Step 3: Substituting $N \to o$

The frequent two-letter word `tN` before `the` and `T` indicates the preposition `to`. Setting $N \to o$ (185 occurrences, 6.8%) resolves `to`, `oC`, `CoI` and `tRo` (Figure 2.6).

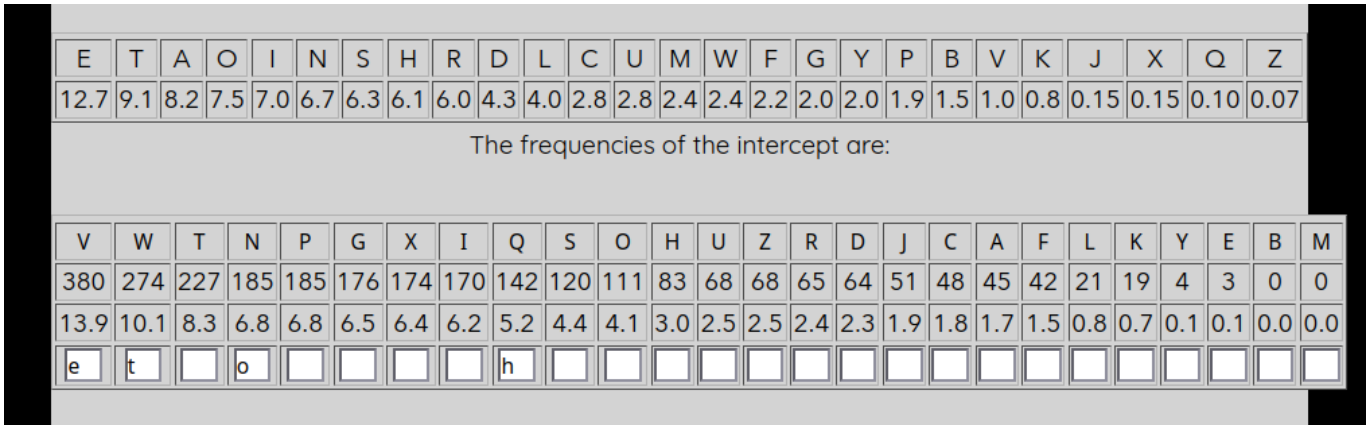


Figure 2.6 – Substitution of $N \to o$ to form `to`

```
Xt ITG RXth TSZoPt DGAeSXeKTASe eCCXHXeGHF. the ATJP oC ZTXS CoI OeSXKeIFthTt ZoIGXGJ
     to the
eZATPPXeP XG KXeGGT ReIe AIoDJht to the ASTHLHhTZAeI eTHh OTF Tt 7 T.Z. theIe the
    SetteIP ReIe
oUeGeO AF ZeStXGJtheXI PeTSP RXth T HTGOSe. the oIOeI oC the SetteIP XG TG eGKeSoUe
    RTPGoteO
TGO the SetteIP JXKeG to T PDAOXIeHtoI. he IeTO theZ TGO oIOeIeOthe XZUoItTGt UTItP
    HoUXeO.
```

2.2.4 Step 4: Substituting $T \to a$

Ciphertext `T` (227 occurrences, 8.3%) appears both as a standalone one-letter word and inside `thTt`, `Tt 7 T.Z.` and `eTHh`. Substituting $T \to a$ yields `a`, `that`, `at` and `eaHh` (Figure 2.7).

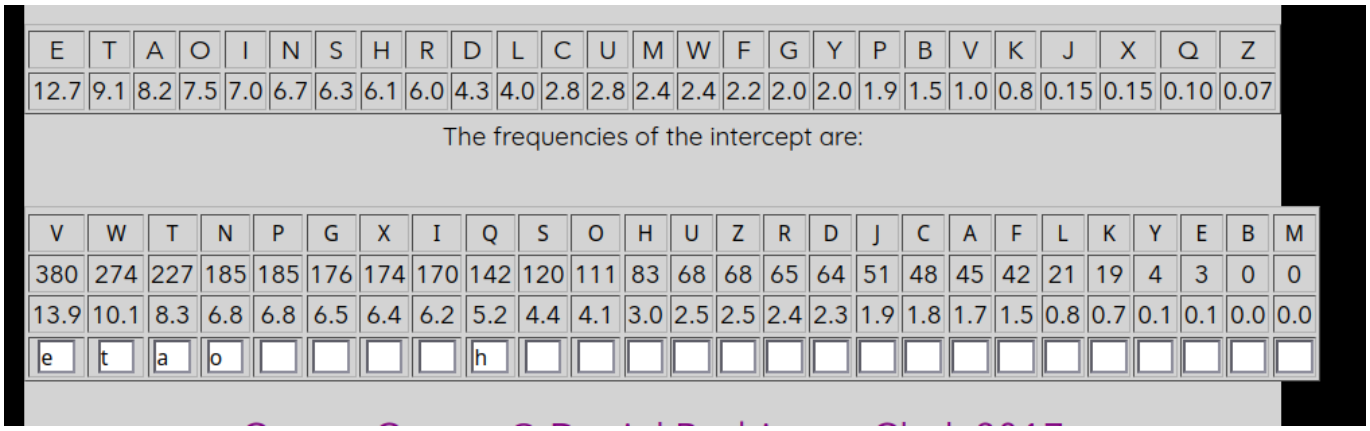


Figure 2.7 – Substitution of $T \to a$

```
Xt IaG RXth aSZoPt DGAeSXeKaASe eCCXHXeGHF. the AaJP oC ZaXS CoI OeSXKeIFthat ZoIGXGJ
     to the
eZAaPPXeP XG KXeGGa ReIe AIoDJht to the ASaHLHhaZAeI eaHh OaF at 7 a.Z. theIe the
    SetteIP ReIe
```

oUeGeO AF ZeStXGJtheXI PeaSP RXth a HaGOSe. the oIOeI oC the SetteIP XG aG eGKeSoUe RaPGoteO
aGO the SetteIP JXKeG to a PDAOXIeHtoI. he IeaO theZ aGO oIOeIeOthe XZUoItaGt UaItP HoUXeO.

2.2.5 Step 5: Substituting $R \to w$

Patterns such as `tRo`, `RXth`, `ReIe`, `RaP` and `RheG` identify $R \to w$ (65 occurrences, 2.4%), producing `two`, `with` (`wXth`), `were` (`weIe`) and `when` (`wheG`) (Figure 2.8).

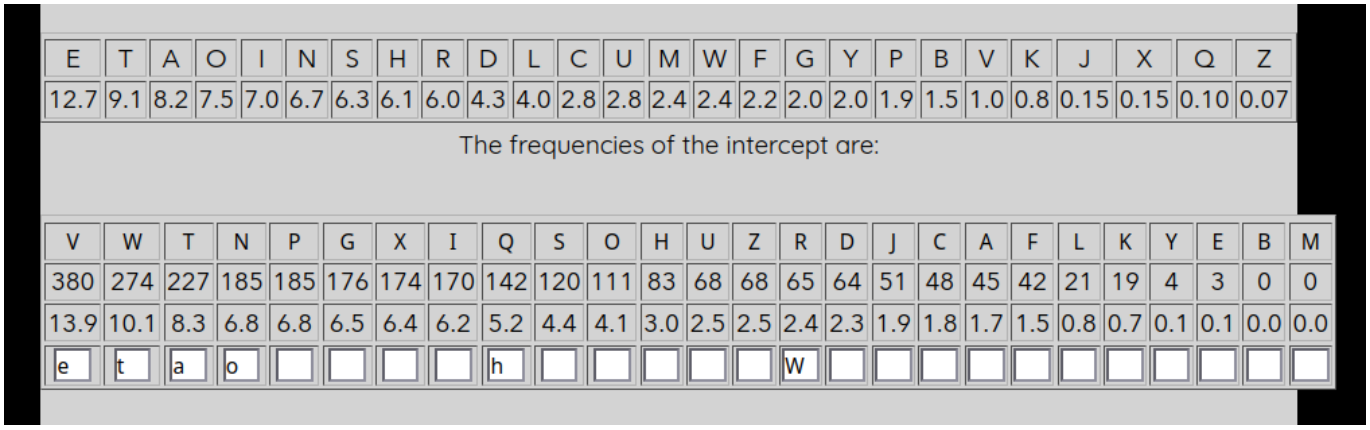


Figure 2.8 – Substitution of $R \to w$ to form words such as `two`

```
Xt IaG wXth aSZoPt DGAeSXeKaASe eCCXHXeGHF. the AaJP oC ZaXS CoI OeSXKeIFthat ZoIGXGJ
     to the
eZAaPPXeP XG KXeGGa weIe AIoDJht to the ASaHLHhaZAeI eaHh OaF at 7 a.Z. theIe the
    SetteIP weIe
oUeGeO AF ZeStXGJtheXI PeaSP wXth a HaGOSe. the oIOeI oC the SetteIP XG aG eGKeSoUe
    waPGoteO
aGO the SetteIP JXKeG to a PDAOXIeHtoI. he IeaO theZ aGO oIOeIeOthe XZUoItaGt UaItP
    HoUXeO.
```

2.2.6 Step 6: Substituting $U \to p$ and $Z \to m$

Time expressions throughout the passage—such as `at 7 a.Z.`, `9:30 a.Z.`, `10 a.Z.`, `2U.Z.`, `7 U.Z.` and `6:30 U.Z.`—immediately determine $Z \to m$ and $U \to p$ (each occurring 68 times, 2.5%), forming `a.m.` and `p.m.` as well as `them` and `opeGeO`.

```
Xt IaG wXth aSmoPt DGAeSXeKaASe eCCXHXeGHF. the AaJP oC maXS CoI OeSXKeIFthat moIGXGJ
     to the
emAaPPXeP XG KXeGGa weIe AIoDJht to the ASaHLHhamAeI eaHh OaF at 7 a.m. theIe the
    SetteIP weIe
opeGeO AF meStXGJtheXI PeaSP wXth a HaGOSe. the oIOeI oC the SetteIP XG aG eGKeSope
    waPGoteO
aGO the SetteIP JXKeG to a PDAOXIeHtoI. he IeaO them aGO oIOeIeOthe XmpoItaGt paItP
    HopXeO.
```

2.2.7 Step 7: Substituting $P \to s$

Short words and word endings such as `waP`, `aP`, `aSmoPt`, `thePe` and `PeaSP` establish $P \to s$ (185 occurrences, 6.8%), giving `was`, `as`, `aSmost` and `these`.

```
Xt IaG wXth aSmost DGAeSXeKaASe eCCXHXeGHF. the AaJs oC maXS CoI OeSXKeIFthat moIGXGJ
     to the
emAassXes XG KXeGGa weIe AIoDJht to the ASaHLHhamAeI eaHh OaF at 7 a.m. theIe the
    SetteIs weIe
```

```
opeGeO AF meStXGJtheXI seaSs wXth a HaGOSe. the oIOeI oC the SetteIs XG aG eGKeSope
    wasGoteO
aGO the SetteIs JXKeG to a sDAOXIeHtoI. he IeaO them aGO oIOeIeOthe XmpoItaGt paIts
    HopXeO.
```

2.2.8 Step 8: Substituting $I \to r$

The pattern `pIopeI` alongside `theIe`, `weIe`, `theXI` and `oIOeI` identifies $I \to r$ (170 occurrences, 6.2%), resolving `proper`, `there` and `were`.

2.2.9 Step 9: Substituting $S \to l$, $C \to f$ and $X \to i$

In the clause `the Setters were retDrGeO to the post oCCXHe`, `Setters` is `letters` ($S \to l$, 120 occurrences, 4.4%) and `post oCCXHe` is `post office` ($C \to f$, 48 occurrences, 1.8%; $X \to i$, 174 occurrences, 6.4%). This simultaneously resolves `it`, `with`, `almost`, `of`, `mail` and `for` (Figure 2.9).

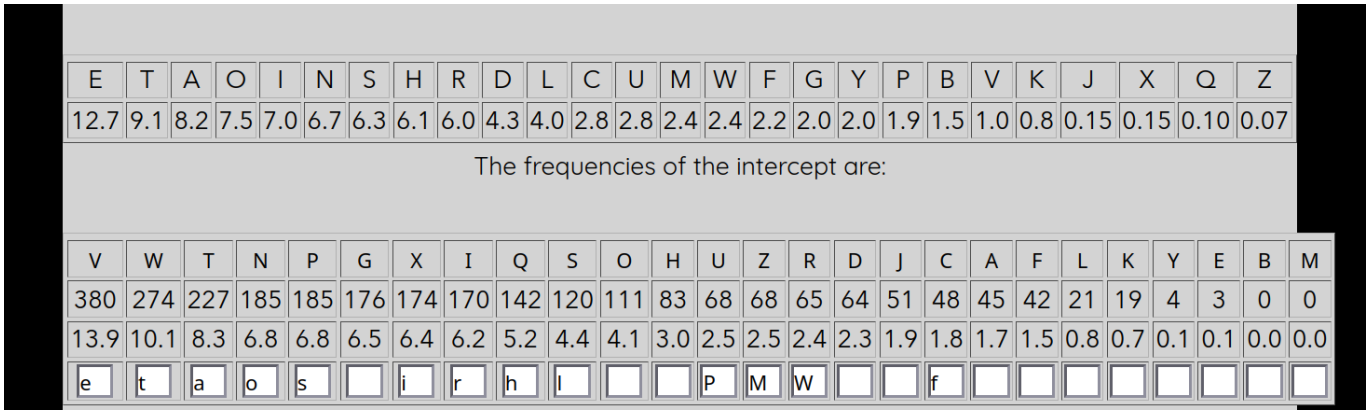


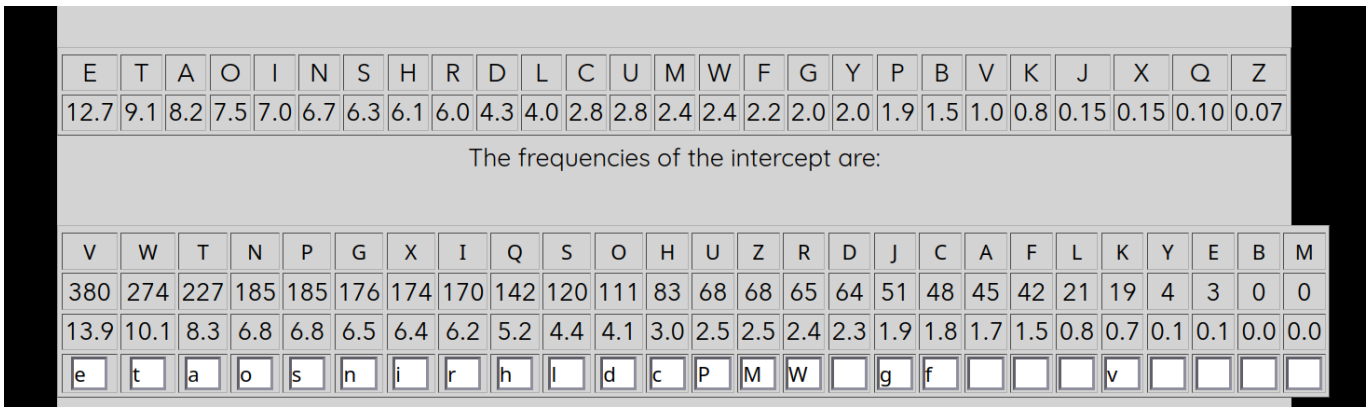
Figure 2.9 – Substitution of $I \to r$, $S \to l$, $C \to f$ and $X \to i$

```
it raG with almost DGAelieKaAle effiHieGHF. the AaJs of mail for OeliKerFthat morGiGJ
     to the
emAassies iG KieGGa were AroDJht to the AlaHLHhamAer eaHh OaF at 7 a.m. there the
    letters were
opeGeO AF meltiGJtheir seals with a HaGOle. the orOer of the letters iG aG eGKelope
    wasGoteO
aGO the letters JiKeG to a sDAOireHtor. he reaO them aGO orOereOthe importaGt parts
    HopieO.
```

2.2.10 Step 10: Substituting $H \to c$, $O \to d$, $G \to n$, $K \to v$ and $J \to g$

The sentence fragment `replaHeO iG their eGKelopes iGtheir oriJiGal orOer aGO the eGKelopes re-sealeO` directly determines five letters: $H \to c$ (83 occurrences, 3.0%), $O \to d$ (111 occurrences, 4.1%), $G \to n$ (176 occurrences, 6.5%), $K \to v$ (19 occurrences, 0.7%) and $J \to g$ (51 occurrences, 1.9%), yielding `replaced in their envelopes intheir original order and the envelopes re-sealed` (Figure 2.10).

```
it ran with almost DnAelievaAle efficiencF. the Aags of mail for deliverFthat morning
     to the
emAassies in vienna were AroDght to the AlacLchamAer each daF at 7 a.m. there the
    letters were
opened AF meltingtheir seals with a candle. the order of the letters in an envelope
    wasnoted
and the letters given to a sDAdirector. he read them and orderedthe important parts
    copied.
```



E	T	A	O	I	N	S	H	R	D	L	C	U	M	W	F	G	Y	P	B	V	K	J	X	Q	Z
12.7	9.1	8.2	7.5	7.0	6.7	6.3	6.1	6.0	4.3	4.0	2.8	2.8	2.4	2.4	2.2	2.0	2.0	1.9	1.5	1.0	0.8	0.15	0.15	0.10	0.07

V	W	T	N	P	G	X	I	Q	S	O	H	U	Z	R	D	J	C	A	F	L	K	Y	E	B	M
380	274	227	185	185	176	174	170	142	120	111	83	68	68	65	64	51	48	45	42	21	19	4	3	0	0
13.9	10.1	8.3	6.8	6.8	6.5	6.4	6.2	5.2	4.4	4.1	3.0	2.5	2.5	2.4	2.3	1.9	1.8	1.7	1.5	0.8	0.7	0.1	0.1	0.0	0.0
e	t	a	o	s	n	i	r	h	l	d	c	P	M	W		g	f				v				

Figure 2.10 – Substitution of $H \to c$, $O \to d$, $G \to n$, $K \to v$ and $J \to g$

2.2.11 Step 11: Substituting $D \rightarrow u$, $A \rightarrow b$, $F \rightarrow y$ and $L \rightarrow k$

From the nearly complete opening lines `it ran with almost DnAelievaAle efficiencF. the Aags of mail for deliverFthat morning to the emAassies in vienna were AroDght to the AlacLchamAer each daF at 7 a.m.`, the remaining four frequent unknowns are identified: $D \rightarrow u$ (64 occurrences, 2.3%), $A \rightarrow b$ (45 occurrences, 1.7%), $F \rightarrow y$ (42 occurrences, 1.5%) and $L \rightarrow k$ (21 occurrences, 0.8%) (Figure 2.11).

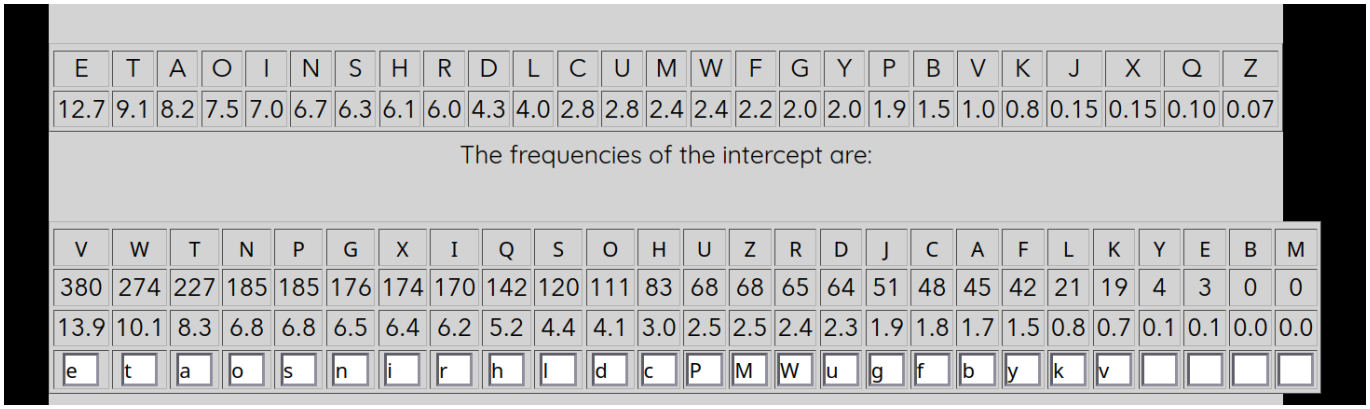


Figure 2.11 – Substitution of $D \rightarrow u$, $A \rightarrow b$, $F \rightarrow y$ and $L \rightarrow k$

```
it ran with almost unbelievable efficiency. the bags of mail for deliverythat morning
    to the
embassies in vienna were brought to the blackchamber each day at 7 a.m. there the
    letters were
opened by meltingtheir seals with a candle. the order of the letters in an envelope
    wasnoted
and the letters given to a subdirector. he read them and orderedthe important parts
    copied.
```

2.2.12 Step 12: Substituting $Y \rightarrow x$ and $E \rightarrow j$

Only two low-frequency symbols remain in the body of the passage:

- ciphertext `Y` (4 occurrences, 0.15%) appears in `for eYample`, `original waY`, `eYcerpted` and `eYistence`, establishing $Y \rightarrow x$;
- ciphertext `E` (3 occurrences, 0.11%) appears in `notEust me`, `subEected` and `enEoyed`, establishing $E \rightarrow j$;

Replacing $Y \rightarrow x$ and $E \rightarrow j$ completes the decryption of all 24 ciphertext symbols present in Variant 12.

3 RESULTS

3.1 Recovered substitution alphabet

Table 3.1 summarizes the complete decryption key derived from the 12 analysis stages.

Table 3.1 – Recovered ciphertext-to-plaintext substitution for Variant 12

Ciphertext	A	B	C	D	E	F	G	H	I	J	K	L	M
Plaintext	b	q*	f	u	j	y	n	c	r	g	v	k	z*
Ciphertext	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Plaintext	o	d	s	h	w	l	a	p	e	t	i	x	m

*B and M do not occur in the ciphertext; $\{B, M\} \to \{q, z\}$ are inferred by elimination.

Every observed ciphertext symbol maps to a distinct plaintext letter. The only unused ciphertext letters are `B` and `M`, and the only unused plaintext letters are `q` and `z`.

3.2 Complete plaintext: reading version

With standard capitalization and spacing restored at line-wrap boundaries, the decrypted Variant 12 passage reads as follows:

It ran with almost unbelievable efficiency. The bags of mail for delivery that morning to the embassies in Vienna were brought to the black chamber each day at 7 a.m. There the letters were opened by melting their seals with a candle. The order of the letters in an envelope was noted and the letters given to a subdirector. He read them and ordered the important parts copied. All the employees could write rapidly, and some knew shorthand. Long letters were dictated to save time, sometimes using four stenographers to a single letter. If a letter was in a language that he did not know, the subdirector gave it to a cabinet employee familiar with it. Two translators were always on hand. All European languages could be read, and when a new one was needed, an official learned it. Armenian, for example, took one cabinet polyglot only a few months to learn, and he was paid the usual 500 florins for his new knowledge. After copying, the letters were replaced in their envelopes in their original order and the envelopes re-sealed, using forged seals to impress the original wax. The letters were returned to the post office by 9:30 a.m.

At 10 a.m., the mail that was passing through this crossroads of the continent arrived and was handled in the same way, though with less hurry because it was in transit. Usually it would be back in the post by 2 p.m., though sometimes it was kept as late as 7 p.m. At 11 a.m., interceptions made by the police for purposes of political surveillance arrived. And at 4 p.m., the couriers brought the letters that the embassies were sending out that day. These were back in the stream of communications by 6:30 p.m. Copied material was handed to the director of the cabinet, who excerpted information of special interest and routed it to the

proper agencies, as police, army, or railway administration, and sent the mass of diplomatic material to the court. All told, the ten-man cabinet handled an average of between 80 and 100 letters a day.

Astonishingly, their nimble fingers hardly ever stuffed letters into the wrong packet, despite the speed with which they worked. In one of the few recorded blunders, an intercepted letter to the Duke of Modena was erroneously re-sealed with the closely similar signet of Parma. When the duke noticed the substitution, he sent it to Parma with the wry note, "Not just me—you too." Both states protested, but the Viennese greeted them with a blank stare, a shrug, and a bland profession of ignorance. Despite this, the existence of the black chamber was well known to the various delegates to the Austrian court, and was even tacitly acknowledged by the Austrians. When the British ambassador complained humorously that he was getting copies instead of his original correspondence, the chancellor replied coolly, "How clumsy these people are!"

Enciphered correspondence was subjected to the usual cryptanalytic sweating process. The Viennese enjoyed remarkable success in this work. The French ambassador, who was apprised of its successes from papers sold him by a masked man on a bridge, remarked in astonishment that "our ciphers of 1200 [groups] hold out only a little while against the ability of the Austrian decipherers." He added that though he suggested new ways of ciphering and continual changes of ciphers, "I still find myself without secure means for the secrets I have to transmit to Constantinople, Stockholm, and St. Petersburg."

CONCLUSIONS

The Variant 12 cryptogram of $2{,}725$ letters is completely decrypted through 12 progressive deduction steps. Relative letter frequencies immediately identify the two most common letters ($V \to e$ at 13.94% and $W \to t$ at 10.06%), which in turn expose high-frequency function words (`the`, `to`, `that`, `with`, `two`) and domain clues such as timestamps (`a.m.`, `p.m.`). Once the core vocabulary is established, longer phrases uniquely determine the remaining medium- and low-frequency letters.

The result confirms that despite the $26! \approx 4.03 \times 10^{26}$ nominal key space of a general monoalphabetic substitution cipher, preserving one-to-one letter correspondence leaves single-letter frequencies, digraphs, trigraphs and word patterns intact, making the cipher vulnerable to statistical and lexical cryptanalysis.

BIBLIOGRAPHY

[1] Gafenco Victor. *Cryptography Laboratory Repository* [online]. GitHub, 2026 [accessed 10.11.2026]. Available: `https://github.com/yoda-dinmd/crypt-labs`

APPENDICES

APPENDIX A

COMPLETE SUPPLIED CIPHERTEXT

The Variant 12 ciphertext below preserves the exact letters, punctuation, missing spaces and line breaks of the intercept.

```
Xw itg rxwq tsznpw dgavsxvktasv vccxhxvghf. Wqv atjp nc ztxs cni ovsxkvifwqtw znigxgj
     wn wqv
vzatppxvp xg Kxvggt rviv aindjqw wn wqv asthlhqtzavi vthq otf tw 7 t.z. Wqviv wqv
    svwwvip rviv
nuvgvo af zvswxgjwqvxi pvtsp rxwq t htgosv. Wqv niovi nc wqv svwwvip xg tg vgkvsnuv
    rtpgnwvo
tgo wqv svwwvip jxkvg wn t pdaoxivhwni. Qv ivto wqvz tgo niovivowqv xzuniwtgw utiwp
    hnuxvo.
Tss wqv vzusnfvvp hndso rixwv ituxosf, tgopnzv lgvr pqniwqtgo. Sngj svwwvip rviv
    oxhwtwvo wn
ptkv wxzv,pnzvwxzvp dpxgj cndi pwvgnjituqvip wn t pxgjsv svwwvi. Xc t svwwvi rtp xg
    tstgjdtjv
wqtw qv oxo gnw lgnr, wqv pdaoxivhwni jtkv xw wn t htaxgvwvzusnfvv ctzxsxti rxwq xw.
    Wrn
witgpstwnip rviv tsrtfp ng qtgo. TssVdinuvtg stgjdtjvp hndso av ivto, tgo rqvg t gvr
    ngv rtp gvvovo,
tgnccxhxts svtigvo xw. Tizvgxtg, cni vytzusv, wnnl ngv htaxgvw unsfjsnw ngsft cvr
    zngwqp wn svtig,
tgo qv rtp utxo wqv dpdts 500 csnixgp cni qxp gvrlgnrsvojv. Tcwvi hnufxgj, wqv
    svwwvip rviv
ivusthvo xg wqvxi vgkvsnuvp xgwqvxi nixjxgts niovi tgo wqv vgkvsnuvp iv-pvtsvo, dpxgj
     cnijvo
pvtsp wnxzuivpp wqv nixjxgts rty. Wqv svwwvip rviv ivwdigvo wn wqv unpw nccxhv af9:30
     t.z.Tw
10 t.z., wqv ztxs wqtw rtp utppxgj wqindjq wqxp hinppintop nc wqvhngwxgvgw tiixkvo
    tgo rtp
qtgosvo xg wqv ptzv rtf, wqndjq rxwq svppqdiif avhtdpv xw rtp xg witgpxw. Dpdtssf xw
    rndso av
athl xg wqv unpw af 2u.z., wqndjq pnzvwxzvp xw rtp lvuw tp stwv tp 7 u.z. Tw 11
t.z.,xgwvihvuwxngp ztov af wqv unsxhv cni udiunpvp nc unsxwxhts pdikvxsstghvtiixkvo.
    Tgo tw 4
u.z., wqv hndixvip aindjqw wqv svwwvip wqtw wqvvzatppxvp rviv pvgoxgj ndw wqtw otf.
    Wqvpv
rviv athl xg wqv pwivtz nchnzzdgxhtwxngp af 6:30 u.z. Hnuxvo ztwvixts rtp qtgovo wn
    wqvoxivhwni
nc wqv htaxgvw, rqn vyhviuwvo xgcniztwxng nc puvhxts xgwvivpw tgoindwvo xw wn wqv
    uinuvi
tjvghxvp, tp unsxhv, tizf, ni itxsrtftozxgxpwitwxng, tgo pvgw wqv ztpp nc oxusnztwxh
    ztwvixts wn
wqv hndiw. Tss wnso, wqv wvg-ztg htaxgvw qtgosvo tg tkvitjv ncavwrvvg 80 tgo 100
    svwwvip t
otf.Tpwngxpqxgjsf, wqvxi gxzasv cxgjvip qtiosf vkvi pwdccvo svwwvip xgwn wqvringj
    uthlvw,
ovpuxwv wqv puvvo rxwq rqxhq wqvf rnilvo. Xg ngv nc wqvcvr ivhniovo asdgovip, tg
    xgwvihvuwvo
svwwvi wn wqv Odlv nc Znovgt rtpviingvndpsf iv-pvtsvo rxwq wqv hsnpvsf pxzxsti pxjgvw
     nc Utizt.
```

Rqvg wqvodlv gnwxhvo wqv pdapwxwdwxng, qv pvgw xw wn Utizt rxwq wqv rif gnwv, " Gnwedpw—
zvfnd wnn." Anwq pwtwvp uinwvpwvo, adw wqv Kxvggvpv jivvwvo wqvzrxwq t astgl pwtiv, t
pqidj, tgo t astgo uincvppxng nc xjgnitghv. Ovpuxwvwqxp, wqv vyxpwvghv nc wqv asthl hqtzavi rtp
rvss lgnrg wn wqv ktixndpovsvjtwvp wn wqv Tdpwixtg hndiw, tgo rtp vkvg wthxwsf thlgnrsvojvo
afwqv Tdpwixtgp. Rqvg wqv Aixwxpq'tzatpptoni hnzustxgvo qdznindpsf wqtw qv rtp jvwwxgj
hnuxvpxgpwvto nc qxp nixjxgts hniivpungovghv, wqv hqtghvssni ivusxvo hnnssf,"Qnr hsdzpf wqvpv
uvnusv tiv!"Vghxuqvivo hniivpungovghv rtp pdaevhwvo wn wqv dpdts hifuwtgtsfwxhprvtwxgj
uinhvpp. Wqv Kxvggvpv vgenfvo ivztiltasv pdhhvpp xg wqxp rnil.Wqv Civghq tzatpptoni, rqn rtp
tuuixpvo nc xwp pdhhvppvp cinz utuvippnso qxz af t ztplvo ztg ng t aixojv, ivztilvo xg tpwngxpqzvgw
wqtw"ndi hxuqvip nc 1200 [jindup] qnso ndw ngsf t sxwwsv rqxsv tjtxgpw wqvtaxsxwf nc wqv
Tdpwixtg ovhxuqvivip." Qv toovo wqtw wqndjq qv pdjjvpwvogvr rtfp nc hxuqvixgj tgo hngwxgdts
hqtgjvp nc hxuqvip, "X pwxss cxgozfpvsc rxwqndw pvhdiv zvtgp cni wqv pvhivwp X qtkv wn
witgpzxw wnHngpwtgwxgnusv, Pwnhlqnsz, tgo Pw. Uvwvipadij."

APPENDIX B

LITERAL DECRYPTED TEXT

The recovered substitution key is applied to every alphabetic character, with plaintext written in lowercase and all original word boundaries retained.

```
it ran with almost unbelievable efficiency. the bags of mail for deliverythat morning
     to the
embassies in vienna were brought to the blackchamber each day at 7 a.m. there the
    letters were
opened by meltingtheir seals with a candle. the order of the letters in an envelope
    wasnoted
and the letters given to a subdirector. he read them and orderedthe important parts
    copied.
all the employees could write rapidly, andsome knew shorthand. long letters were
    dictated to
save time,sometimes using four stenographers to a single letter. if a letter was in
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    two
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    letters were
replaced in their envelopes intheir original order and the envelopes re-sealed, using
     forged
seals toimpress the original wax. the letters were returned to the post office by9:30
     a.m.at
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    would be
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    and at 4
p.m., the couriers brought the letters that theembassies were sending out that day.
    these
were back in the stream ofcommunications by 6:30 p.m. copied material was handed to
    thedirector
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    proper
agencies, as police, army, or railwayadministration, and sent the mass of diplomatic
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```

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